

Bridging the Gap: Sustainable Collaboration between CG Production and Educational Institutions

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ABSTRACT

One of the major challenges that new employees in Japan's CG entertainment industry faces is the skill gap between the industry and academia. Recognizing the urgent need for an effective approach to bridge this gap, this study with a new initiative with Nihon Kogakuin College, proposes and implements an innovative educational model, drawing analogies from the "Melting Pot" and the "Salad Bowl." The "Salad Bowl" model honors the uniqueness and needs of both participating companies and students, emphasizing the preservation of diversity. Through its implementation, it has been elucidated that this new educational approach brings about an improvement in student skills and high satisfaction levels, paving the way for success in future workplace environments through real-world industry experience. Additionally, it has been effective in resolving recruitment challenges of smaller studios, overcoming faculty shortages, providing a concrete evaluation of soft skills, and creating employment opportunities tailored to each company. These investigative outcomes underscore the significance and effectiveness of establishing a sustainable collaboration between industry and academia, realizing the provision of education programs that cater to the diverse industrial demands of Japan.

CCS CONCEPTS

• applied computing; education; interactive learning environments; collaborative learning; • social and professional topics; professional topics; computing profession; employment issues;

KEYWORDS

education-industry collaboration, skill gap, sustainable education program, diversified needs, and skills

ACM Reference Format:

Harutaka Matsunaga, Yukari Nagai, and Kazunori Miyata. 2023. Bridging the Gap: Sustainable Collaboration between CG Production and Educational Institutions. In *SIGGRAPH Asia 2023 Educator's Forum (SA Educators Forum '23)*, December 12–15, 2023, Sydney, NSW, Australia. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3610540.3627003>

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SA Educators Forum '23, December 12–15, 2023, Sydney, NSW, Australia

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ACM ISBN 979-8-4007-0311-9/23/12.

<https://doi.org/10.1145/3610540.3627003>

1 INTRODUCTION

In Japan, a multitude of specialized vocational schools offer CG, Animation, and Game Development education, which seek to foster close collaborations with the industry. Concurrently, a significant number of emergent studios, often led by young founders, are burgeoning, each nurturing its own ambitions and cultural identities. These studios, alongside well-established animation companies and major corporations, contribute to a diverse industry landscape, each holding its unique culture and aspirations. Employees tend to stay in one company, and each company has distinctive cultures and unique workflows. On the other hand, the rapid pace of technological advancements necessitates that these specialized schools continuously integrate the latest techniques from the CG industry, highlighting the urgent and ongoing need for synchrony between the skills demanded in the industry and those provided in the educational arena. However, a prevailing issue has been the inadequacy of traditional educational curricula in these specialized schools in keeping pace with the swift technological evolution and the varying demands of the diverse industry landscape. The conventional curricula have tended to skew towards generic software proficiency and basic training, thereby failing to nurture an understanding of distinct studio production cultures and the development of soft skills. Moreover, in emergent studios, evaluating the soft skills of prospective interns, while keeping educational costs in check, has proven to be quite challenging.

2 PROPOSING A NEW EDUCATIONAL MODEL

Vocational schools fundamentally assume the role of nurturing talents for the industrial sector. However, the traditional industry-academia collaborative curriculum has been integrating diverse demands from various companies into a single program, adopting a generalized approach. As a result, crucial elements forming corporate culture such as unspoken knowledge and organizational memory were overlooked. The process of averaging the demands from companies by vocational schools tended to result in curricula that don't fully align with any particular corporate culture. This approach proved to be ineffective, especially in a milieu like Japan where diverse studio cultures and unique workflows coexist.

To address this challenge, we propose a new educational model termed the "Salad Bowl" model. This model reinterprets the metaphors of the salad bowl and melting pot, originally used in the sociological context to depict the immigrant society of the United States, within the ambit of CG education [Civic Issues, 2019]. Contrasting the traditional educational model, likened to a "Melting Pot" that melds diverse corporate demands into a single curriculum, the proposed "Salad Bowl" model emphasizes coexistence while retaining the diversity of corporate cultures.

The new educational model centers on recognizing and appreciating the unique identities and requirements of both students and companies. Analogous to the components of a salad, each participant retains their individuality while collaboratively contributing to co-creation. A key component of this paradigm is the application of the theory of "Legitimate Peripheral Participation" [Lave and Wenger, 1991]. This theory fosters a cooperative learning environment where novices assimilate knowledge through interactions with professionals. Our approach ensures that students are exposed to the latest industry trends and companies' unique tacit knowledge, elements that are elusive in a standardized curriculum.

This initiative was launched in 2019 at the CG Department of Nihon Kogakuin College, fostering partnerships with five pioneering companies in the CG, game, and VFX industries. As of now, 13 dedicated instructors and 42 program participants are actively engaged, ensuring the program's continued growth and influence.

3 RESULTS

The average satisfaction score for the program was 9.29 out of a maximum of 10, with a standard deviation of 1.16. On the other hand, the average skill improvement score stood at 8.25 with a standard deviation of 1.94. The correlation coefficient between these two was 0.49, indicating a moderate positive relationship.

A distinctive feature contributing to the sustainability of this program is the return of alumni as instructors. This strategy not

only addresses potential instructor shortages but also embeds a profound sense of loyalty and affiliation among both the returning graduates and the current students.

4 CONCLUSION

For the program to truly achieve its intended success, it's imperative that companies clearly recognize the benefits they can derive from it and actively participate. Strengthening collaborations with graduates and fostering partnerships with productions for regional contributions will be key. In the future context of CG education, it's anticipated that there will be a shift in focus, not just towards CG technology but also towards cultivating soft skills such as communication abilities. This would enhance students' self-affirmation and further solidify the bonds between academia and the industry. Going forward, it will be crucial to further deepen these initiatives and strive for holistic invigoration.

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